Sensitive Attribute Reconstruction from a Rashomon Set of Decision Trees

1 Method

1.1 Rashomon set construction

We binarize a curated subset of the UCI Adult dataset: the sensitive target T = `sex` (single indicator column), background-knowledge attribute A = `marital-status` (collapsed to 4 categories: Married, Never-married, Divorced, Widowed-Separated), background-knowledge attribute B = `education` (collapsed to 5 categories: Below-HS, HS-grad, Some-college, Bachelors, Grad+), plus `age` and `hours-per-week` median-split indicators and a `capital-gain`> 0 indicator – 13 binary columns in total. The prediction label Y is income (> \$50K vs. not).

This curated feature set is a deliberate scope reduction from the full ∼60-column binarization of Adult. On the full feature set, TreeFARMS (Xin et al., 2022), at a regularization loose enough to build a Rashomon set within a few minutes on a laptop, only produced trees shallow enough (1–2 splits) that none of them ever split on `sex`: every background-knowledge condition was then identical by construction, since the released trees carried no information about T at all. Reducing the competing feature count lets T, A, and B actually get selected as splits while keeping the exhaustive Rashomon-set search tractable on consumer hardware.

For each of 10 repetitions (seeds), we sample $n = 100$ rows from Adult and build a Rashomon set with TreeFARMS using regularization $\lambda = 0.02$ and Rashomon bound multiplier 0.2 (i.e. all trees within 20% of the optimal objective are kept). This reliably produces Rashomon sets of 100+ trees in under a second.

1.2 Reconstruction attack

We reimplement the constraint-programming reconstruction attack of Ferry et al. (2024, *"Trained Random Forests Completely Reveal your Dataset"*, known as DRAFT), adapted to read TreeFARMS's native tree representation and to support fixing a subset of feature columns to their true value for every row (the attacker's background knowledge). The attack is solved

with OR-Tools CP-SAT (no commercial solver license required): one boolean variable per (row, feature), one-hot-sum constraints per categorical attribute, and per-tree, per-leaf constraints forcing the solved table to reproduce the exact per-class training-example counts of the released trees. The label Y is assumed known to the attacker in every condition, consistent with this literature: the released trees predict Y, so Y is treated as public.

For a released ensemble of the first m trees from the Rashomon set (TreeFARMS's enumeration order; no diversity-aware selection strategy is applied), we solve the attack under four background-knowledge conditions, holding the data sample and released trees fixed within a repetition:

- **None**: neither A nor B is known.
- **A**: A is known for every row; B is not.
- **B**: B is known for every row; A is not.
- **A+B**: both A and B are known.

In every condition T remains unknown. We sweep $m \in \{1, 5, 10, 25, 50\}$.

1.3 Evaluation

The solver has no notion of row identity beyond Y (fixed per row index) and whichever columns are pinned as known; under partial background knowledge, rows sharing a label are otherwise interchangeable as far as the model's constraints are concerned. Before scoring T, we therefore match each reconstructed row to a true row via the Hungarian algorithm (within each Y-label group), using Hamming distance over every column *except* T as the matching cost – excluding T from the matching cost avoids letting the evaluator's own knowledge of the true values inflate T's measured accuracy. Reconstruction error is the fraction of matched rows whose reconstructed T disagrees with the truth.

We compare against two baselines: a uniform-random guess, and a holdout-marginal guess that samples T from its empirical marginal estimated on a 2000-row holdout sample disjoint from the attacked 100 rows and never passed to the Rashomon-set construction or the attack.

2 Results

Figure 1 shows the mean reconstruction error on T across the 10 repetitions, for each background-knowledge condition and number of released trees m, against the two baselines (means taken over all repetitions, since the baselines do not depend on m).

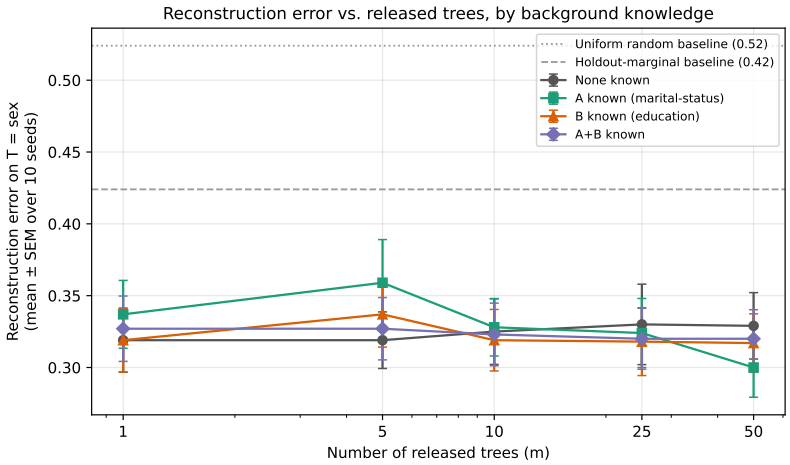


Figure 1: Reconstruction error on $T = \texttt{sex}$ vs. number of released trees m, by background-knowledge condition. Error bars are ± 1 standard error of the mean over 10 repetitions.

The released trees leak substantially more than either baseline. All four conditions solve to error around 0.30–0.36 across every m tested, against a uniform-random baseline of 0.52 and a holdout-marginal baseline of 0.42. This confirms the core premise motivating this project: a released Rashomon set of trees leaks information about a sensitive attribute T beyond what is available from chance or from knowing only the population-level marginal of T – even though none of T's values are ever given directly to the attacker.

Background knowledge of A/B did not produce a clear, consistent effect at this scale. The four conditions track each other closely at every m, with error bars overlapping throughout; there is no m at which any condition is clearly and consistently separated from the others. A paired, per-repetition comparison (fraction of the 10 seeds where a condition's error is strictly lower than the *None* condition's error, for the same sample and released trees) does show a directionally consistent trend as m grows:

	$m = 1$	$m = 5$	$m = 10$	$m = 25$	$m = 50$
A beats None	0.0	0.0	0.2	0.4	0.6
B beats None	0.1	0.1	0.3	0.4	0.5
A+B beats None	0.1	0.1	0.3	0.3	0.6

At $m = 1$ or $m = 5$, background knowledge essentially never helps (and in the case of A at $m = 5$, the mean error is visibly *higher* than with no

background knowledge at all). From $m = 10$ onward, the fraction of repetitions where background knowledge helps rises toward, but does not clearly exceed, one half. With only 10 repetitions, none of these differences should be read as statistically established; the monotonic-in-m pattern across all three knowledge conditions is the most we are willing to claim.

Why background knowledge might help only weakly here. The attack has no objective function beyond feasibility: OR-Tools CP-SAT returns *some* table consistent with the released leaf counts, not a calibrated or most-likely one. Fixing A or B to their true values can only propagate into a tighter constraint on T through a tree whose root-to-leaf path splits on *both* T-related and A/B-related features together. Because our regularization keeps trees shallow (1–3 splits each, to keep the Rashomon-set search tractable on a laptop), such joint paths are relatively rare, which plausibly explains why the effect of background knowledge only begins to show up once enough trees ($m \gtrsim 25$) are released for joint paths to accumulate across the ensemble.

Limitations. This is an MVP-scale result, not a conclusive one: (i) the 13-column curated feature set and shallow trees were a scope decision made for laptop tractability and should be revisited with deeper trees and/or more compute if time allows; (ii) released trees are the first m in TreeFARMS's enumeration order, not selected by a diversity-aware strategy (cf. Hsu et al., 2026, who show selection strategy materially affects leakage); (iii) only 10 repetitions were run, and no formal significance test is reported; (iv) the solver's parallel search (OR-Tools default) means repeated runs with the same seed are not bit-for-bit reproducible, only distributionally comparable; (v) no calibration or confidence measure is reported – only point reconstruction error.